RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 27 August 2026, 8:51 AM
State	Finished
Completed on	Thursday, 17 September 2026, 9:41 AM
Time taken	21 days
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Divide and Conquer function to find the floor of x
int findFloor(int nums[], int low, int high, int x) {
    // Base case: If search space is exhausted
    if (low > high) {
        return -1;
    }

    // If element is greater than or equal to all elements in the current range
    if (x >= nums[high]) {
        return nums[high];
    }

    // Divide step: find the middle index
    int mid = low + (high - low) / 2;

    // If exact match found
    if (nums[mid] == x) {
        return nums[mid];
    }

    // If x is smaller than nums[mid], the floor must lie in the left half
    if (x < nums[mid]) {
        return findFloor(nums, low, mid - 1, x);
    }

    // If x is greater than nums[mid], nums[mid] is a candidate.
    // Check if a better floor exists in the right half.
    int rightFloor = findFloor(nums, mid + 1, high, x);

    if (rightFloor != -1) {
        return rightFloor;
    } else {
        return nums[mid];
    }
}

int main() {
    int n;
    scanf("%d", &n);

    int nums[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }

    int x;
    scanf("%d", &x);

    int result = findFloor(nums, 0, n - 1, x);

```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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